

Unit 1 : Basic of AI

Topics:

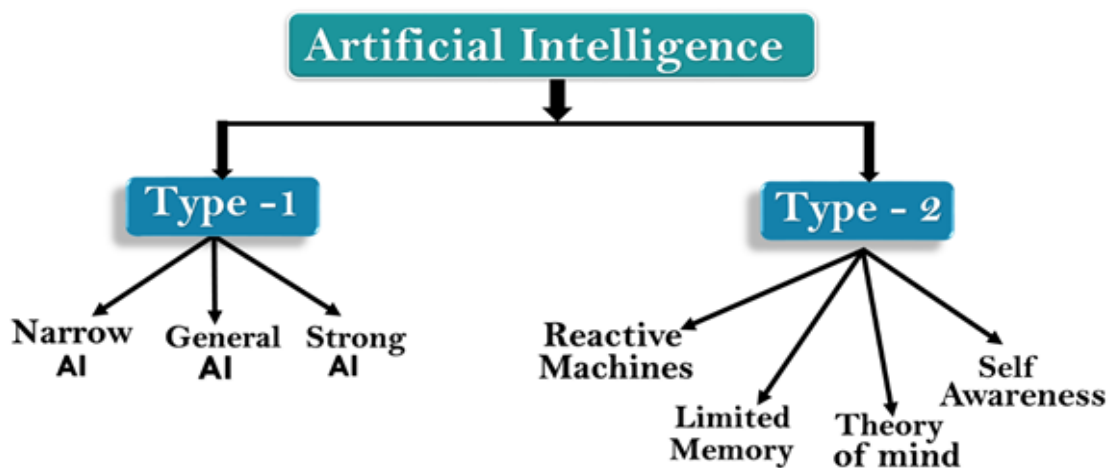
- Definition: Artificial Intelligence
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❖ Definition: Artificial Intelligence

- Artificial Intelligence is a way of making a computer, a computer-controlled robot, or a software think intelligently, in the similar manner the intelligent humans think.
- Artificial intelligence is the study of how to make computers do things which, at moment people do better.
- Artificial Intelligence is composed of two words **Artificial** and **Intelligence**, where Artificial defines "man-made," and intelligence defines "thinking power", hence AI means "a man-made thinking power."
- So, we can define AI as:- **"It is a branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions"**
- Artificial Intelligence exists when a machine can have human based skills such as learning, reasoning, and solving problems

❖ Types of Artificial Intelligence

- Artificial Intelligence can be divided into various types, there are mainly two types of main categorization which are based on capabilities and based on functionality of AI. Following is a flow diagram which explains the types of AI.



- **AI type-1: Based on Capabilities**

- 1. Weak AI or Narrow AI:**

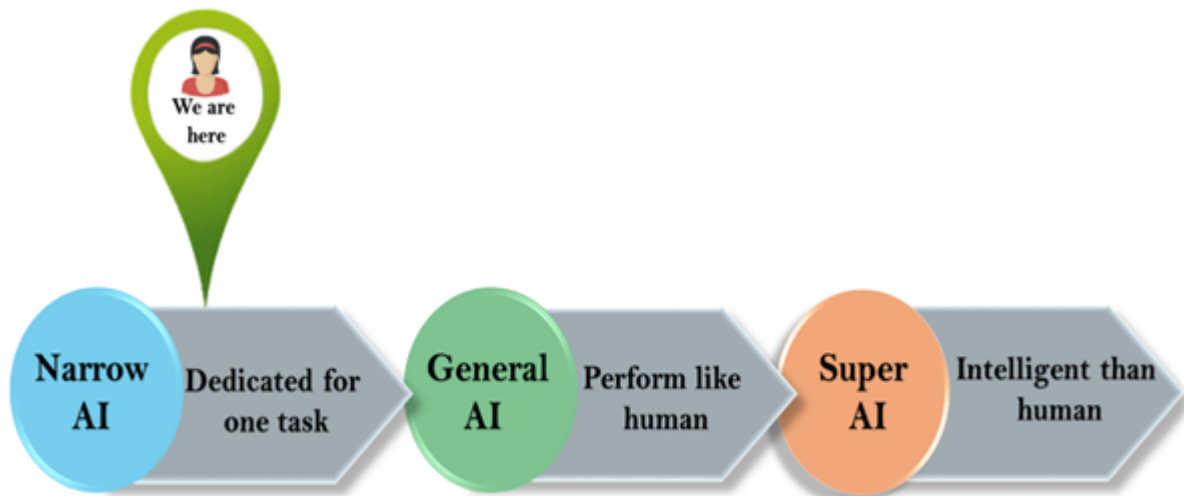
- Narrow AI is a type of AI which is **able to perform a dedicated task with intelligence**. The most common and currently available AI is Narrow AI in the world of Artificial Intelligence.
 - Narrow AI **cannot perform beyond its field or limitations**, as it is only trained for one specific task. Hence it is also termed as weak AI. Narrow AI can fail in unpredictable ways if it goes beyond its limits.
 - **Apple Siri** is a good example of Narrow AI, but it operates with a limited pre-defined range of functions.
 - **IBM's Watson supercomputer** also comes under Narrow AI, as it uses an Expert system approach combined with Machine learning and natural language processing.
 - Some Examples of Narrow AI are playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition.

- 2. General AI:**

- General AI is a type of intelligence which **could perform any intellectual task with efficiency like a human**.
 - The **idea** behind the general AI is to make such a **system which could be smarter and think like a human by its own**.
 - Currently, there is **no such system exist** which could come under general AI and can perform any task as perfect as a human.
 - The worldwide researchers are now focused on developing machines with General AI.
 - As systems with general AI are still under research, and it will take lots of efforts and time to develop such systems.

- 3. Super AI:**

- Super AI is a level of Intelligence of Systems **at which machines could surpass human intelligence, and can perform any task better than human with cognitive properties**. It is an **outcome of general AI**.
 - Some key characteristics of strong AI include the ability to think, to reason, solve the puzzle, make judgments, plan, learn, and communicate **by its own**.
 - Super AI is still a hypothetical concept of Artificial Intelligence. Development of such systems in real is still world changing task.



- **Artificial Intelligence type-2: Based on functionality**

- 1. **Reactive Machines**

- Purely reactive machines are the **most basic** types of Artificial Intelligence.
 - Such AI systems **do not store memories or past experiences** for future actions.
 - These machines only **focus on current scenarios and react on it** as per possible best action.
 - **IBM's Deep Blue system** is an example of reactive machines.
 - **Google's AlphaGo** is also an example of reactive machines.

- 2. **Limited Memory**

- Limited memory machines **can store past experiences or some data for a short period of time.**
 - These machines **can use stored data for a limited time period only.**
 - **Self-driving cars** are one of the best examples of Limited Memory systems. These cars can store the recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

- 3. **Theory of Mind**

- Theory of Mind AI should **understand human emotions, people, beliefs, and be able to interact socially like humans.**
 - These types of AI machines are **still not developed**, but researchers are making lots of efforts and improvements for developing such AI machines.

- 4. **Self-Awareness**

- Self-awareness AI **is the future** of Artificial Intelligence. These machines **will be super intelligent, and will have their own consciousness, sentiments, and self-awareness.**
 - These machines **will be smarter than the human mind.**
 - Self-Awareness AI does not exist in reality still and it is a hypothetical concept.

❖ Evolving Stages of Artificial Intelligence

Stage 1- Rule Bases System

In this stage, AI surrounds us in everything from Robot Process Automation used in business to autopilots used in aircraft.

Stage 2- Context-awareness and Retention

The machines based on algorithms are trained on the best human knowledge and experience with knowledge based on specific domains. The existing knowledge base can be updated once new queries and solutions arise. The most common application to be Robo-advisors or chatbots which can be used in a forward customer inquiry that can help human in everything.

Stage 3- Domain-specific aptitude

In this stage, the advanced systems can develop mastery in particular domains. Due to the capability to store and process a wide volume of information to make decisions the capabilities of these machines can be more than humans. Cancer diagnosis is one of the applications and Google Deepmind's system that defeated 18 times world go champion Lee Sadol proved the domain-specific expertise.

Stage 4- Reasoning systems

This machine algorithm has a concept of intellect, intentions, knowledge, and their own logic. They have the capacity to reason, interact and, deal with other machines and humans too. Such algorithms will be in the commercial arena in some years.

Stage 5- Artificial General Intelligence

This is the main objective of many scientists working in the field of AI- to develop a machine with intelligence like humans. The artificial general intelligence or self-aware systems are not in application to date but not many years left to discover this avenue. This is the stage represented in many sci-fiction movies that machines are leading humans in intelligence.

Stage 6- Artificial Super Intelligence(ASI)

This stage is one notch of the Artificial General Intelligence, the algorithms will be capable of defeating smartest of the humans in all the domains. This stage can be the real solution to the problems that are still complex for the human mind like poverty, hunger, and climate change. This will be the stage where machines made by humans will outsmart humans too.

Stage 7- Singularity and excellency

The development path led by ASI will lead us to the point in time where the capabilities of humans will see the tremendous extension. With the help of this ASI, technology humans would be able to connect the minds of each other and be like the human internet. The technology would be much like an imaginary life where ideas, and thoughts could be shared by just one flick of mind. Humans would be capable of connecting with other forms of life like animals, plants, and other natural activities. While some argue that humans will never reach such levels and some say that it is feasible and with this exponential growth rate would be achieved somewhere in 2045.

❖ The AI Problems

- Much of the early work in the field of AI focused on formal tasks, such as game playing and theorem proving.
- Game playing and theorem proving share the property that people who do them well are considered to be displaying *Intelligence*.
- Initially computers could perform well at those tasks simply by being fast at exploring a large number of solution paths and then selecting the best one.
- Humans learn mundane (ordinary) tasks since their birth. They learn by perception, speaking, using language, and training. They learn Formal Tasks and Expert Tasks later.
- Another early foray into AI focused on commonsense reasoning, which includes reasoning about physical objects and their relationship to each other, as well as reasoning about actions and their consequences.
- As AI research progressed, techniques for handling large amount of world knowledge were developed.
- New tasks reasonably attempted such as perception, natural language understanding and problem solving in specialized domains.
- Some of the task domains of artificial intelligence are presented in table I.
- Earlier, all work of AI was concentrated in the mundane task domain.

Mundane tasks	Formal tasks	Expert tasks
Perception – Computer Vision – Speech, Voice	Games – Go – Chess (Deep Blue) – Ckeckers	Engineering – Design – Fault Finding – Manufacturing – Monitoring
Natural Language Processing – Understanding – Language Generation – Language Translation	Mathematics – Geometry – Logic – Integration and Differentiation	Scientific Analysis
Common Sense Reasoning	Theorem Proving	Financial Analysis
Planning		Medical Diagnosis
Robot Control		

Task Domains of AI

- Later, it turned out that the machine requires more knowledge, complex knowledge representation, and complicated algorithms for handling mundane tasks.
- This is the reason why AI work is more flourishing in the Expert Tasks domain now, as the expert task domain needs expert knowledge without common sense, which can be easier to represent and handle.

❖ Difference - Artificial Intelligence vs. Machine Learning

ARTIFICIAL INTELLIGENCE	MACHINE LEARNING
AI stands for Artificial intelligence, where intelligence is defined acquisition of knowledge intelligence is defined as the ability to acquire and apply knowledge.	ML stands for Machine Learning which is defined as the acquisition of knowledge or skill
The aim is to increase the chance of success and not accuracy	The aim is to increase accuracy, but it does not care about success
AI is aiming to develop an intelligent system capable of performing a variety of complex jobs.	Machine learning is attempting to construct machines that can only accomplish the jobs for which they have been trained.
It works as a computer program that does smart work.	Here, the tasks systems machine takes data and learns from data.
The goal is to simulate natural intelligence to solve complex problems.	The goal is to learn from data on certain task to maximize the performance on that task.
AI has a very broad variety of applications.	The scope of machine learning is constrained.
AI is decision-making.	ML allows systems to learn new things from data.
It is developing a system that mimics humans to solve problems.	It involves creating self-learning algorithms.
AI will go for finding the optimal solution.	ML will go for a solution whether it is optimal or not.
AI leads to intelligence or wisdom.	ML leads to knowledge.
AI is a broader family consisting of ML and DL as its components.	ML is a subset of AI.
Three broad categories of AI are : 1. Artificial Narrow Intelligence (ANI) 2. Artificial General Intelligence (AGI) 3. Artificial Super Intelligence (ASI)	Three broad categories of ML are : 1. Supervised Learning 2. Unsupervised Learning 3. Reinforcement Learning
AI can work with structured, semi-structured, and unstructured data.	ML can work with only structured and semi-structured data.

AI's key uses include- ● Siri, customer service via catboats ● Expert Systems ● Machine Translation like Google Translate ● Intelligent humanoid robots such as Sophia, and so on.	The most common uses of machine learning- ● Facebook's automatic friend suggestions ● Google's search algorithms ● Banking fraud analysis ● Stock price forecast ● Online recommender systems, and so on.
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❖ Application of Artificial Intelligence

- AI has been dominant in various fields such as –
 1. Gaming – AI plays vital role in strategic games such as chess, poker, tic-tac-toe, etc., where machine can think of large number of possible positions based on heuristic knowledge.
 2. Natural Language Processing – It is possible to interact with the computer that understands natural language spoken by humans.
 3. Expert Systems – There are some applications which integrate machine, software, and special information to impart reasoning and advising. They provide explanation and advice to the users.
 4. Computer Vision Systems – These systems understand, interpret, and comprehend visual input on the computer.
 5. Speech Recognition – Some intelligent systems are capable of hearing and comprehending the language in terms of sentences and their meanings while a human talks to it. It can handle different accents, slang words, noise in the background, change in human's noise, etc.
 6. Handwriting Recognition – The handwriting recognition software reads the text written on paper by a pen or on screen by a stylus. It can recognize the shapes of the letters and convert it into editable text.
 7. Intelligent Robots – Robots are able to perform the tasks given by a human. They have sensors to detect physical data from the real world such as light, heat, temperature, movement, sound, bump, and pressure. They have efficient processors, multiple sensors and huge memory, to exhibit intelligence. In addition, they are capable of learning from their mistakes and they can adapt to the new environment.



practical

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<https://sites.google.com/view/ait-ai/Home/practicals>

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Types of ai-

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